The Files You Will Use

File	Type · CRS as stored	Task	What it is
Task1_Overlay/focus_nlcd2021_30m.tif	raster, 30 m · EPSG:5070	1	NLCD 2021 land cover codes; forest = 41, 42, 43
Task1_Overlay/padus41_mthood_gorge.shp	3,043 polygons · EPSG:4326	1	Protected areas; field GAP_Sts = 1–4 (text)
Task1_Overlay/focus_counties_pnw.shp	6 polygons · EPSG:4269	1	Study-area outline
Task2_Buffer/coxsbazar_camps_2023.shp	33 polygons · EPSG:32646	2	Camp outlines (camp level only)
Task2_Buffer/ucdp_ged_2023_near_camps.csv	410 rows, longitude/latitude	2	Conflict events 2023; deaths_best, where_prec
Task3_Image_Classification/summer_2023/ (4 files)	1 band each, 10 m · EPSG:5070	3	Sentinel-2 Jun–Sep 2023: B02, B03, B04, B08 (× 10,000)
Task3_Image_Classification/spring_2023/ (4 files)	1 band each, 10 m · EPSG:5070	3	Sentinel-2 Apr–May 2023, same bands and grid
Task3_.../champaign_training_rois.gpkg	154 squares · EPSG:5070	3	Training areas, 5 classes, SCP import fields
Task3_.../champaign_validation_rois.gpkg	120 squares · EPSG:5070	3	Separate check areas for Accuracy
Task3_.../champaign_nlcd2021_30m.tif	raster, 30 m · EPSG:5070	3	NLCD 2021 for the same area (reference)
Extensions/ A_Spatial_Statistics · C_OSINT	counties · UCDP 2023 + countries	opt.	Optional work after class (see README.md)

Keep every .csvt next to its .csv — it tells QGIS which columns are numbers. The README.md in the root folder lists every file.

Task 1 · Overlay: How Much Forest Is Protected? (1/2)

Question In six counties around Mount Hood (Oregon/Washington), what share of the forest lies inside protected areas — and how does the answer change with the definition of “protected”?

1 **Load and set the CRS.** Add focus_nlcd2021_30m.tif, padus41_mthood_gorge.shp, focus_counties_pnw.shp. Project ▸ Properties ▸ CRS ▸ EPSG:5070.

2 **Make a forest mask.** Raster ▸ Raster Calculator: ("focus_nlcd2021_30m@1" >= 41) AND ("focus_nlcd2021_30m@1" <= 43) → forest_mask.tif (1 = forest, 0 = not).

3 **Turn it into polygons.** Raster ▸ Conversion ▸ Polygonize (Raster to Vector) → field DN. Then Processing ▸ Extract by attribute: DN = 1 → forest.

4 **Prepare the protected layer.** Processing ▸ Reproject layer (PAD-US → EPSG:5070) ▸ Fix geometries ▸ Extract by expression: "GAP_Sts" IN ('1','2') — GAP_Sts is text, so use quotes.

5 **Dissolve it.** Vector ▸ Geoprocessing Tools ▸ Dissolve (no field) → protected_12. PAD-US records overlap; without Dissolve the area is counted twice.

What GAP status means

1 wilderness, left to natural processes

2 managed for biodiversity (refuges, wild rivers)

3 protected from development but open to logging, grazing, mining — most national-forest land

4 no known protection mandate

Strict definition = GAP 1–2. Broad definition = GAP 1–3. You will compute both.

Slow computer? First Raster ▸ Extraction ▸ Clip Raster by Mask Layer to one county (e.g. Hood River) and run the steps on that county only.

Task 1 · Overlay: Measure the Gap (2/2)

6 Protected forest. Vector ▸ Geoprocessing Tools ▸ Intersection: input forest, overlay protected_12 → forest_protected.

7 The gap. Vector ▸ Geoprocessing Tools ▸ Difference: input forest, overlay protected_12 → forest_gap.

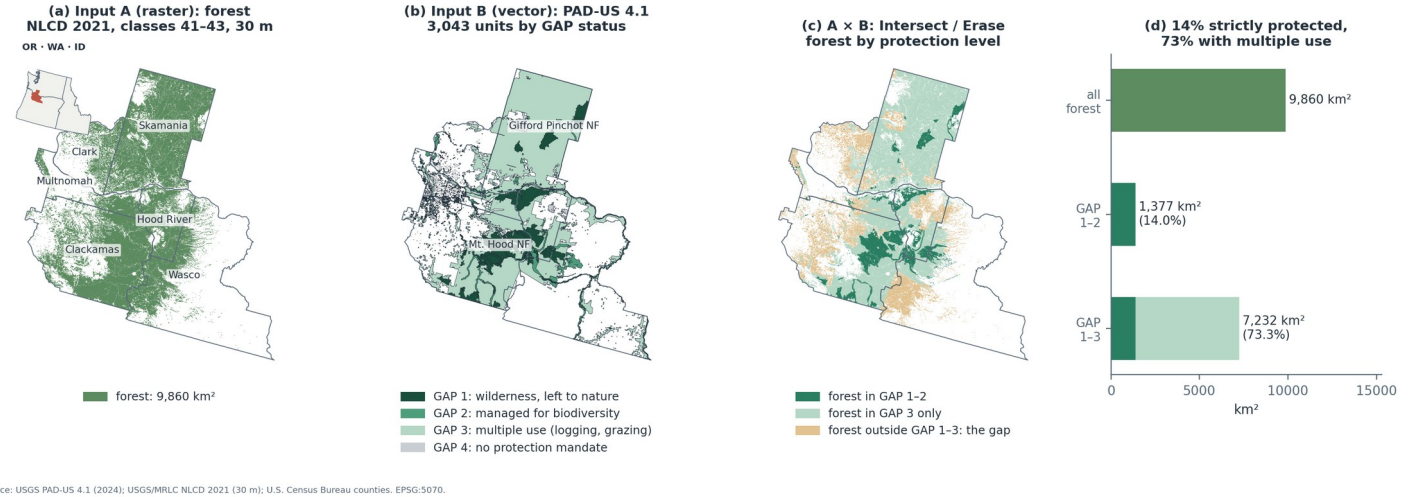
8 Measure. Open the Field Calculator on each layer, new decimal field km2 = area($geometry) / 1e6. Then Vector ▸ Analysis Tools ▸ Basic Statistics for Fields → SUM of km2.

9 Repeat with GAP 1–3. Steps 4–7 with "GAP_Sts" IN ('1','2','3'). Write both answers down.

Self-check (whole focus area)

All forest ≈ 9,860 km² · inside GAP 1–2 ≈ 1,377 km² (14.0 %) · inside GAP 1–3 ≈ 7,232 km² (73.3 %). Differences under 1 % are fine. A total far above 9,860 km² means you skipped Dissolve.

For the memo: which definition of "protected" do you use, and why?



What the result looks like (notebook figure): (a) forest, (b) PAD-US by GAP status, (c) the overlay, (d) the areas.

TASK 2 · 15–30′

Task 2 · Buffer: Which Refugee Camps Are Near Conflict?

1 New project, CRS EPSG:32646. Add coxsbazar_camps_2023.shp (33 camp outlines, already in UTM 46N, metres).

2 Add the events as points. Layer ▸ Add Layer ▸ Add Delimited Text Layer: ucdp_ged_2023_near_camps.csv, X field = longitude, Y field = latitude, CRS EPSG:4326.

3 Reproject the events. Processing ▸ Reproject layer → EPSG:32646. Buffering in degrees gives wrong distances.

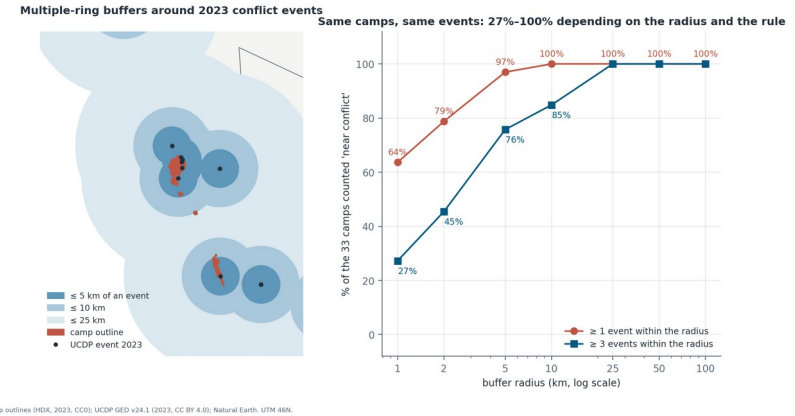
4 Buffer the events. Vector ▸ Geoprocessing Tools ▸ Buffer, distance 5000 m, tick Dissolve result → buf_5km. Repeat for 1 km and 10 km.

5 Count camps inside. Vector ▸ Research Tools ▸ Select by Location: select camps that intersect buf_5km. The number selected appears at the top of the attribute table.

6 Stricter rule (≥ 3 events). Buffer the camps by 5 km → Vector ▸ Analysis Tools ▸ Count Points in Polygon (points = events) → select NUMPOINTS >= 3.

Radius	Camps with ≥ 1 event	Camps with ≥ 3 events
1 km	21 of 33 (64 %)	9 (27 %)
2 km	26 (79 %)	15 (45 %)
5 km	32 (97 %)	25 (76 %)
10 km	33 (100 %)	28 (85 %)
25 km	33 (100 %)	33 (100 %)

Self-check: your counts should match this table.



Ethics: these are camp outlines only. Never map individual shelters or blocks.

For the memo: the answer runs from 27 % to 100 % depending on the radius and the rule. Pick one, say why, and show the range.

Task 3 · Classify a Satellite Image with SCP (1/2)

Question Can a random forest map land cover in Champaign–Urbana from a Sentinel-2 image — and how much does a second date help? Training and check areas are ready-made, so you only press the buttons and read the results.

1 **Open SCP.** Install the Semi-Automatic Classification Plugin before class (Plugins ▸ Manage and Install Plugins; random forest also needs scikit-learn — see the SCP installation page).

2 **Band set 1 = summer.** SCP ▸ Band set ▸ open file ▸ select the four files in Task3_Image_Classification/summer_2023/ (B02 blue, B03 green, B04 red, B08 near-infrared). In the working toolbar set RGB = 3-2-1 for natural colour, 4-3-2 for false colour.

3 **Load the training areas.** SCP ▸ Training input ▸ new file training.scpx ▸ Import vector: champaign_training_rois.gpkg. Match the fields macroclass_id, macroclass_name, class_id, class_name ▸ Import. You get 154 squares in 5 classes.

4 **Run 1.** SCP ▸ Classification ▸ input band set 1 ▸ tick Use Macroclass ID ▸ Algorithm: Random forest, number of trees 100 ▸ Run → class_summer.tif.

5 **Check it.** SCP ▸ Postprocessing ▸ Accuracy: classification = class_summer.tif, reference = champaign_validation_rois.gpkg, field macroclass_id ▸ Run. Write down the overall accuracy.

What goes into the model

Input (features): for every 10 m pixel, its reflectance in each band of the band set — 4 numbers in Run 1, 8 in Run 2.

Labels: the class of each training square: 1 Water · 2 Developed · 3 Forest · 4 Grass and pasture · 5 Cropland. Each square is 90 × 90 m (81 pixels) where NLCD 2021 shows one class only.

Check areas: 120 different squares, at least 300 m from any training square, used only for the accuracy report.

The model learns which reflectance goes with which class, then predicts every pixel of the image.

Task 3 · Add a Second Date and Compare (2/2)

6 **Band set 2 = summer + spring.** In the Band set tab add a new band set (+) and load the four summer files and the four files in spring_2023/ (8 bands).

7 **Run 2.** Classification with input band set 2, same settings → class_summer_spring.tif.

8 **Check it.** Postprocessing ▸ Accuracy with the same validation file. Compare with Run 1.

Self-check	Run 1 · summer	Run 2 · + spring
Overall accuracy	≈ 0.66	≈ 0.81
Water (recall)	0.65	0.79
Developed	0.76	0.87
Forest	0.68	0.71
Grass and pasture	0.62	0.74
Cropland	0.56	0.88

Random forests are random: ± 0.02 between runs is normal.

Why the second date helps

The summer image is a June–September median. In mid-summer, corn and soybean fields are dense green canopy and look almost like forest in these four bands: forest NDVI 0.83, cropland 0.80. Run 1 confuses them.

In April–May the fields are still bare soil while forest and grass are green. Adding that date lets the model separate them: cropland recall rises from 0.56 to 0.88.

Time is a feature. Choosing when the image was taken is as much an analytical decision as choosing the algorithm.

For the memo: report per-class accuracy, not just the overall number, and name the class your map gets wrong most often.

The labels come from NLCD 2021; the image is from 2023, so some “errors” are real change or label mistakes.

Task 4 · Make the Map and Write the Memo

The map (Project ▸ New Print Layout)

Check (lecture slide 5)	Ask yourself
1 Purpose	Can a reader say in one sentence what the map shows?
2 Visual variable	Darker = more? Categories in different hues?
3 Classification	Which method and how many classes? Stated?
4 Colour	Colour-blind safe; readable in greyscale?
5 Data integrity	Rates not counts; equal-area CRS; source and date on the map
6 Essentials only	Title, legend, scale, source; nothing decorative

Export: Layout ▸ Export as Image (PNG, 300 dpi) or Export as PDF. One map per analysis; at most two maps.

The memo (200–300 words)

Question — one sentence: what policy question does your analysis answer?

Data and units — layers, CRS, and the definitions you chose (GAP 1–2 or 1–3; radius; rule).

Operations and results — the tools in order, with your numbers.

One policy implication — what should change, for whom, on the strength of this result?

What it cannot establish — e.g. designation is not enforcement; straight-line distance is not travel distance; the map shows where, not why.

AI and ethics (optional) — did an AI suggestion help? Does any output locate vulnerable people precisely?

A template with these headings is in Task4_Memo/week8_memo_template.md.